

A Teacher annotation (offline, small subset)



Offline only

Stratified subset S
small sample of posts

Teacher discarded

Few-shot prompt
rubric + strict schema

Frozen LLM teacher
DeepSeek/GPT-5

$q(x') \in [0, 1]^K$
concept probabilities

Teacher outputs discarded after distillation training

Supervisions

B Student training and inference (textless)

Textless hashed input

Hashed features $\varphi(x')$
char 3-5-grams, signed



Logistic students s_c
one per concept c
learns from $q(x')$

Distillation loss
 $\text{BCE}(\hat{h} q_c, q_c) + \ell_2$

$\hat{h} q_c(x')$
student predictions

User-level aggregation
MEAN·VAR·MAX·ACF₁